CYB 305 | DIGITAL FORENSICS AND INVESTIGATION METHODS

LABORATORY PRACTICAL 01

Preliminary forensic examination of suspected payment fraud
Week 1: Computer Forensics Fundamentals | Year 3 undergraduate

Duration	60 minutes	Delivery	Microsoft Teams
Case	CYB305-001 (fictional)	Tools	CyberChef; optional LibreOffice Calc

1. Purpose and learning outcomes

You will examine a set of synthetic digital records associated with a suspected unauthorised change to a university tuition portal. This preliminary examination focuses on file integrity, evidence correlation, time normalisation, provenance, analytical uncertainty and documentation. You will work with supplied records rather than acquire a forensic image of a device.

By the end of this practical, you should be able to:

1. Verify file integrity using SHA-256, document a discrepancy and explain its investigative significance.
2. Correlate authentication, portal, web and email artefacts using timestamps and record identifiers.
3. Normalise UTC records to West Africa Time (WAT, UTC+01:00) and reconstruct an evidence-based sequence of events.
4. Distinguish recorded activity, reasonable inference, competing explanations and unsupported claims.
5. Identify evidence-handling defects, propose preservation measures and describe a relevant legal or ethical safeguard.

2. Case brief

At 09:37 WAT on 12 October 2026, a student at fictional Westbridge University reported an unexpected tuition payment email. ICT staff subsequently found that a beneficiary reference in the tuition portal may have changed during a scheduled maintenance window. A preliminary set of exported records was supplied for examination. The change may involve an unauthorised action, an error or activity for which the authorisation record is incomplete. Your task is to determine what the records support and identify what cannot yet be established.

Scope. Examine only the supplied synthetic case materials. Do not contact any website or IP address appearing in the records. Do not assume that an account, session, IP address or sender name identifies a particular individual. No transaction ledger has been supplied.

3. Resources and evidence organisation

Before the session, download the case evidence ZIP and retain the original ZIP unchanged. The archive contains SOURCE_EVIDENCE, WORKING_COPIES, a SHA-256 source manifest, a report template and a README. Treat the supplied source evidence as read-only and conduct your examination on verified working copies.

Tools. Use the open-source CyberChef application at https://gchq.github.io/CyberChef/ (SHA2 set to 256 bits). LibreOffice Calc at https://www.libreoffice.org/ is optional for sorting and filtering CSV files. You can read the remaining files in a plain text editor. No paid software is required.

4. Session plan

Time	Task	Your output
00-05	Read case brief; define investigative question	A focused investigation objective
05-15	Examine source and working-copy integrity	SHA-256 comparison and discrepancy note
15-35	Correlate records and compare configuration	Observations supported by cited artefacts
35-45	Reconstruct a normalised chronology	Timeline citing evidence and event IDs
45-55	Evaluate findings and evidence handling	Qualified conclusion and preservation advice
55-60	Submit preliminary report	One to two pages

5. Practical tasks

Task A. Evidence receipt and integrity (05-15 minutes)

1. Record the received archive filename, time of receipt and case ID. Do not modify files in SOURCE_EVIDENCE.
2. In CyberChef, use SHA2 with 256 bits to hash the actual file contents of working-copy files E05 and E06. Do not retype their contents. Compare each result with the corresponding source SHA-256 value in SHA256_SOURCE_MANIFEST.csv.
3. Record which files match and which do not. If a working copy does not match, calculate the hash of the corresponding file in SOURCE_EVIDENCE, record the discrepancy, then make and verify a fresh working copy before relying on it.

Question A. What can a hash mismatch establish, and what cannot be concluded from it?

Task B. Examine and correlate digital records (15-35 minutes)

1. Read E07 and identify the approved change, its conditions and the maintenance window.
2. Compare source-verified E05 and E06. Identify the fields that changed and distinguish the portal banner from the beneficiary reference.
3. Examine E02, E03 and E04. Find events associated with payment-setting changes. Correlate records using the account, session_id, request_id and timestamp. Cite the event IDs supporting your observations.
4. Examine E01 and E08 for records associated with the reported email. Consider timing and sender details, but distinguish a temporal association from proof of a common operator.

Question B. Does the change request cover every recorded modification? What additional evidence would help resolve uncertainty about authorisation?

Task C. Reconstruct a chronology (35-45 minutes)

1. Convert UTC timestamps in E02 and E04 to WAT by adding one hour. E03, E08 and E09 already use WAT.
2. Construct a chronology containing at least four significant events. Cite the file and event ID for each event and include corroboration from another artefact where available.
3. State any limitation of timestamps, session IDs or network addresses that affects interpretation or attribution.

Task D. Evaluate findings and evidence handling (45-55 minutes)

1. State the strongest conclusion supported by the records, and explain at least one alternative explanation or unresolved possibility.
2. Examine E09 for an incomplete evidence transfer or missing receipt acknowledgement. Explain why the documentation gap matters and what clarification is needed.
3. Recommend one immediate preservation measure, one forensic readiness improvement and one Nigerian legal or ethical safeguard relevant to this case.

Task E. Submit your report (55-60 minutes)

Prepare a one- to two-page preliminary forensic examination report using STUDENT_REPORT_TEMPLATE.txt or the same headings in your own document. Your report must contain two hash comparisons, at least three cited technical observations, a WAT chronology with at least four entries, a reasoned finding with a limitation or alternative explanation, and a custody or preservation recommendation. Submit your report before the session ends.

6. Assessment (20 marks)

Criterion	Marks
File integrity, verification and discrepancy response	4
Technical examination and corroboration of artefacts	6
Chronology and correct time-zone handling	4
Critical evaluation, alternative explanations and limits	4
Documentation and practical recommendations	2
Total	**20**

7. Professional and legal considerations

The Evidence Act 2011 (as amended), the Cybercrimes (Prohibition, Prevention, etc.) Act 2015 (as amended), and the Nigeria Data Protection Act 2023 provide context for discussing the collection, use and disclosure of electronic records in Nigeria. You are not required to decide whether an exhibit would be admissible in court. Explain why technical integrity verification alone cannot establish admissibility, authorship or the lawfulness of acquisition. All case information is fictional.

8. Access and submission guidance

You can complete the examination using the downloaded case files and CyberChef. If you have limited internet access, obtain an offline copy of CyberChef before the session and use it with the same evidence package. You do not need to share your screen to complete the individual report. Your submission should present what the evidence establishes, what remains uncertain and which additional records should be preserved.

Relevant lecture topics: Digital evidence and preservation (slides 11-21); forensic readiness (23-25); investigator responsibilities (27-30); legal compliance (33-34).